

Executive Summary

Employee Attrition Prediction Project

Project Objective

The objective of this project was to understand the factors influencing employee attrition and identify practical ways to improve employee retention. By analyzing employee data, the project aimed to provide HR with insights that support better workforce planning and reduce employee turnover.

Key Findings

- Employee attrition is relatively low overall; however, employees who leave show clear and identifiable patterns.
- The **Sales Department** experiences the highest employee turnover and should be prioritized for retention initiatives.
- Employees reporting **low job satisfaction** are significantly more likely to resign.
- **Work-life balance** has a strong impact on employee retention, with poor balance increasing the likelihood of attrition.
- Salary influences employee decisions, but employee engagement, career growth, and workplace satisfaction are equally important.
- Employees in their **early years of service** are more likely to leave, highlighting the importance of onboarding and early career support.

Business Value

Three predictive models were evaluated during the analysis. The **Random Forest model** delivered the most reliable results for identifying employees who may be at risk of leaving. This allows HR teams to take proactive measures before valuable employees resign.

Recommendations

- Focus retention efforts on the Sales department.
- Conduct regular employee satisfaction surveys and address concerns promptly.
- Improve work-life balance through flexible policies and workload management.
- Provide clear career development and internal growth opportunities.
- Strengthen engagement programs for newly hired employees.
- Use predictive analytics regularly to identify employees who may require additional support.

Conclusion

This project demonstrates that employee attrition is driven by multiple workplace factors rather than salary alone. By improving employee satisfaction, supporting career development, promoting work-life balance, and using predictive insights, the organization can reduce turnover, retain talented employees, and build a stronger, more productive workforce.